www.tntextbooks.in

Other Sugar Compounds

Other Polysaccharides	Structure	Functions
Inulin	Polymer of fructose	It is not metabolised in the human body and is readily filtered through the kidney
Hyaluronic acid	Heteropolymer of d glucuronic acid and D-N acetyl glucosamine	It accounts for the toughness and flexibility of cartilage and tendon
Agar	Mucopolysaccharide from red algae	Used as solidifying agent in culture medium in laboratory
Heparin	Glycosamino glycan contains variably sulphated disaccharide unit present in liver	Used as an anticoagulant
Chondroitin sulphate	Sulphated glycosaminoglycan composed of altering sugars (N-acetylglucosamine and glucuronic acid)	Dietery supplement for treatment of osteoarthritis
Keratan sulphate	Sulphated glycosaminoglycan and is a structural carbohydrate	Acts as cushion to absorb mechanical shock



Human can't digest cellulose but herbivores can digest them with the help of bacteria present in the gut which produces enzyme **cellulase.** This is an example of **mutualism.**

8.4 Lipids

The term lipid is derived from *greek* word lipos, meaning fat. These substances are not soluble in polar solvent such as water but dissolve in non-polar solvents such as benzene, ether, chloroform. This is because they contain long hydrocarbon chains that are non-polar and thus hydrophobic. The main groups of compounds classified as lipids are triglycerides, phospholipids, steroids and waxes.

8.4.1 Triglycerides

Triglycerides are composed of single molecule of glycerol bound to 3 fatty acids. These include fats and oils. Fatty acids are long chain hydrocarbons with a carboxyl group at one end which binds to one of the hydroxyl groups of glycerol, thus forming an ester bond. Fatty acids are structural unit of lipids and are carboxylic acid of long chain hydrocarbons. The hydrocarbon can vary in length from 4 – 24 carbons and the fat may be saturated or unsaturated. In saturated fatty acids the hydrocarbon chain is single bonded (Eg. palmitic acid, stearic acid) and in unsaturated fatty acids (Eg. Oleic acid,